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SIXTH SEMESTER **B. Tech. (SE)**

END SEMESTER EXAMINATION ***May-2019***

SE-314 ARTIFICIAL INTELLIGENCE

Time: 3:00 Hours *Max. Marks: 70*

Note:	Answer any ***ALL*** questions. Assume suitable missing data, if any.

1 Answer all the following questions: [3.5x4=14]

[a] Describe Turing test for intelligence machine. Also, give major characteristics of intelligent machine.

[b] Explain Bayes rule with an example.

[c] What is meant by intelligent program? Explain with example.

[d] What is heuristic search?

2 Answer any TWO of the following questions:

[a] Describe Breadth First Search (BFS) and compare with Depth First Search (BFS). [7]

[b] What is meant by state-space search? Explain it using water jug problem. [7]

[c] Explain hill climbing search algorithm and compare with steepest hill climbing search algorithm. [7]

3 Answer any TWO of the following questions:

[a] What are the merits and demerits of semantic nets and frames? [7]

[b] State and explain A* algorithm. [7]

[c] Define fuzzy sets and compare with classical sets. Also comment how it is different from probability theory. [7]

4 Answer any TWO of the following questions:

[a] Explain the working of the artificial neural network (ANN) with the help of a neat diagram. [7]

[b] How a Neural network is for Learning? Explain with example. [7]

[c] What is meant by bi-directional search? Explain its advantages. [7]

5 Answer any TWO of the following questions:

[a] Explain Genetic Algorithm with an example. [7]

[b] Draw the architecture of Expert system defining functionality of different components. [7]

[c] State and explain an evolutionary algorithm for optimal search. [7]

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